EPROMFOUNDRY INC.
VINTAGE SIGNAL PROCESSORS · REISSUED IN LOGIC

DATA SHEET

G2 AMIGA TRM
Revision 1.0

Mike Wolak · mike@epromfoundry.com · (512) 944-8586

G2 Amiga release 1.0 · 2026-11-03

G2 Amiga Technical Reference

The structure of the G2 Amiga: the Amiga core and the RISC-V SoC on one Spartan-6, their clocks and memory in DDR2, the register page through which the firmware serves the core, the boot chain, every buffer, and the measured performance of release 1.0.

HIGHLIGHTS

- An A1200-class Amiga (TG68K 68020, AGA) and a RISC-V SoC on one XC6SLX100.
- Amiga memory in two DDR2 chips behind fabric caches.
- One register page carries disks, input, picture, sound, network and serial.
- Firmware from the stick: a flash write only for a new fabric.

KEY PARAMETERS

PARAMETER	VALUE
FPGA	XC6SLX100-2FGG484
DDR2	2 × 64 MB
Core clock	28.636364 MHz
SoC clock	50 MHz
Register page	0x8000_E000
Flash	M25P64, 8 MB

1 ARCHITECTURE

One XC6SLX100 holds two machines. The Amiga core runs on a gated copy of the 50 MHz system clock at the Amiga's own rate; its memory lives in DDR2 behind caches. A VexiiRiscv RISC-V SoC on the same fabric runs the firmware from DDR2: it owns the USB host, the stick, Ethernet, the video heads, the codec and the setup screen, and it serves the core's disk sectors, input, sound, network frames and serial bytes through one register page.

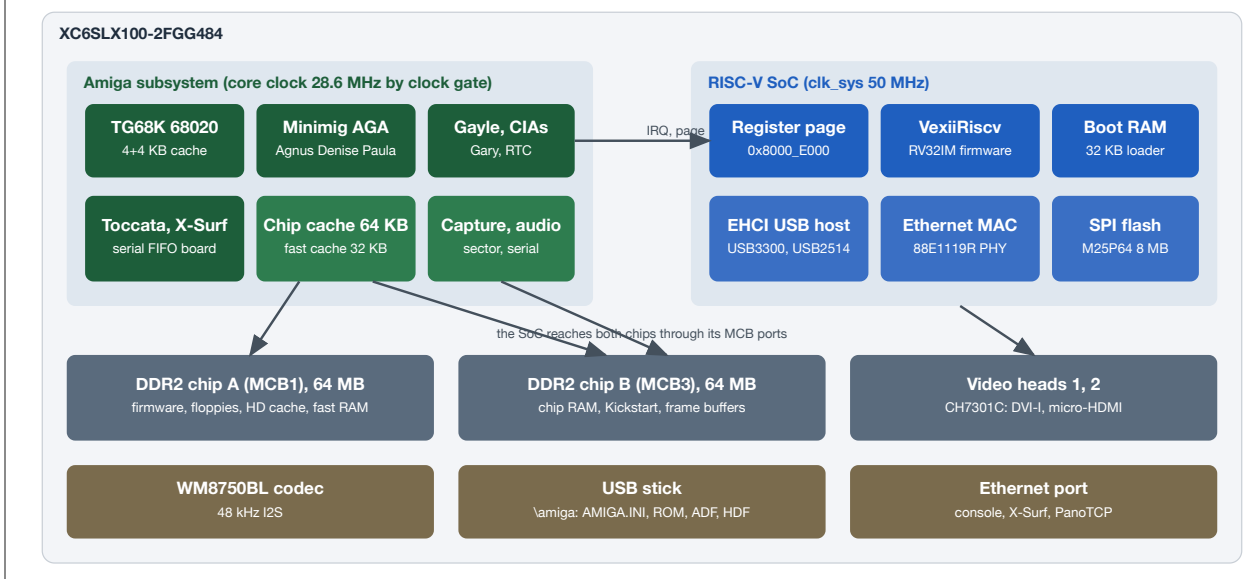


Figure 1. Blocks of the G2 Amiga fabric and the board parts they drive.

BLOCK	IMPLEMENTATION	ORIGIN
Chipset	Minimig AGA: Agnus, Denise, Paula, CIA A/B, Gary, Gayle; OCS, ECS and AGA selectable	NanoMig (MiSTle-Dev, Till Harbaum), Minimig lineage (Dennis van Weeren, Rok Krajnc, MiSTer contributors), GPL-3.0
CPU	TG68K 68000/68010/68020 with a 2-way 4 KB instruction and 4 KB data cache (early hit, two core clocks)	Tobias Gubener (TG68K); Rok Krajnc, Alexey Melnikov (cache)
Sound	Paula; Toccata (AD1848 model)	ranzbak fpga-toccata, BSD-2-Clause
Network	X-Surf 100 (NE2000 model) with a host mailbox	Minimig-AGA Ethernet branch
Serial	Paula's UART bridge; serial FIFO board (Zorro II, 0x0A1C:1)	Mike Wolak, EPROM Foundry Inc.
SoC	VexiiRiscv RV32IM with caches, MCB DDR2 ports, video heads, codec path, Ethernet MAC, SPI flash with a write guard	VexiiRiscv (Charles Papon, MIT); Mike Wolak, EPROM Foundry Inc.
USB host	xps_usb_host v1.01.a EHCI on the USB3300 ULPI PHY and USB2514 hub	Xilinx, Inc. (licensed EDK core, not in the public source)

2 CLOCKS

CLOCK	FREQUENCY	SOURCE AND USE
clk_sys	50.000 MHz	PLL from the PHY's 125 MHz: SoC, MCB user ports, register page
Amiga core clock	28.636364 MHz NTSC, 28.375160 MHz PAL (mean)	clk_sys through a BUFGCE: an NCO adds NCO_INC (0x929E_4149 NTSC, 0x9147_E3C4 PAL) every clk_sys cycle; each carry owes the core one edge
DDR2	125 MHz (250 MHz MCB sysclk_2x)	Two Spartan-6 MCBs, CL3
ULPI	60 MHz	From the USB3300
USB, codec reference	24.000 MHz	Fabric PLL 125 MHz × 24 / 5 / 25
Video	per mode	A DCM reprogrammed for each head mode

The clock gate withholds core edges while a chip-memory access is not yet confirmed by its cache; withheld edges are owed and repaid at up to one per clk_sys cycle, so the long-term rate is the NCO rate and the machine's own timing is unchanged. With sdc_hold the gate also holds the core while a disk request waits for the firmware, so a sector arrives at once in the core's time.

3 MEMORY

3.1 DDR2 layout

RANGE (CPU ADDRESS)	SIZE	CONTENT
0x1000_0000–0x10FF_FFFF	16 MB	Chip A: the firmware (AMIGA.FW loads at 0x1000_0000), its data and heap to 0x1200_0000
0x1100_0000	8 MB max	Chip A: staging buffer of the loader for AMIGA.FW

RANGE (CPU ADDRESS)	SIZE	CONTENT
0x1200_0000–0x127F_FFFF	4 × 2 MB	Chip A: the floppy images DF0–DF3, held whole
0x1280_0000–0x12FF_FFFF	8 MB	Chip A: the hard-disk cache, 128 blocks of 64 KB
0x1300_0000–0x137F_FFFF	8 MB	Chip A: Amiga fast RAM (one Zorro II board at $200000), with the Kickstart copy for turbo Kickstart at fast-RAM byte 0x78_0000
0x1460_0000	2 × 803 KB	Chip B: the text console's two buffers (setup screen), head 2 mirrored 16 MB higher
0x1480_0000–0x14AF_FFFF	3 × 1 MB	Chip B: the captured Amiga frames; head 2's copy at 0x1580_0000
0x1600_0000–0x16FF_FFFF	16 MB	Chip B: the Amiga's chip-bus space: chip RAM from 0, slow RAM, Kickstart at +0x78_0000 (two bytes of each word exchanged)

Bit 31 of a CPU address selects the uncached alias (0x9000_0000 chip A, 0x9400_0000 chip B); every buffer the core or a video head reads is written through it.

3.2 Amiga memory

AMIGA RANGE	SIZE	BACKING
$000000	512 KB–2 MB chip (CHIP=)	Chip B behind the chip cache (2-way, 64 KB, 64-byte lines)
$C00000	0–1.5 MB slow (SLOW=)	Chip B, chip bus
$200000	0, 2, 4 or 8 MB fast (FAST=)	Chip A behind the fast-RAM cache (2-way, 32 KB)
$F80000	256 or 512 KB Kickstart	Chip B, read only to the core
$DC0000	MSM6242 registers	Loaded by the firmware from network time

4 THE RISC-V TO AMIGA INTERFACE

The firmware reaches the Amiga subsystem through one 4 KB register page at 0x8000_E000 (clk_sys domain, one-clock acknowledge) and interrupt line 6 of the SoC. Events cross into the core's clock domain as toggles.

OFFSET	ACCESS	REGISTER	FUNCTION
0x000	R	ID	0x414D_4947 ("AMIG")
0x004	RW	CTRL	run[0], core_reset[1], sdc_hold[2], serial IRQ enable[3], mirror[5], frame IRQ enable[6], audio IRQ enable[7]; reset 0x6
0x008	R	STATUS	chip_flush_busy[0], field[8], lace[9], head_sel[16]
0x00C	W	CACHE	invalidate[0], flush[1], audio clear[3] (with run 0)
0x020, 0x024	RW	NCO_INC, AUD_INC	Core clock and 48 kHz audio sample rate in core time
0x02C, 0x030	RW	FB_BASE, VID	Frame-buffer base in chip B; capture hofs[26:16], width[10:0]
0x034, 0x038	R	FRAMES, FRAME_INFO	Frames completed; {fast[26], lines[25:16], lace[2], buffer[1:0]}

OFFSET	ACCESS	REGISTER	FUNCTION
0x03C	RW	IRQ	Pending {serial[3], mcu[2], audio[1], frame[0]}
0x04C–0x054	R/RW	AUD_DATA, AUD_LEVEL, AUD_WMARK	Stereo sample pop; FIFO level and overflow; watermark (reset 256)
0x058–0x078	R	Counters	Chip and fast cache misses and write-backs, held clocks, lost and owed edges
0x070	RW	DISP	The frame buffer the heads scan
0x080	RW	MCU_TX	Companion stream: W {start[16], target[9:8], byte[7:0]}; R {busy[31], last byte}
0x090–0x0A4	R/W	SDC_REQ, SECTOR, GO, DATA, MOUNT, SIZE	The sector port (4.2)
0x0A8–0x0B4	RW	MBX_ADDR, MBX_DATA, XS_MAC	X-Surf host mailbox (16-bit words, auto-increment) and station address
0x0B8, 0x0BC	RW/R	VID_FAST, LINE_DROPS	Capture offset of 31 kHz lines; lines lost
0x0C0–0x0CC	R/W	MIDI_RX, TX, ST, SERPER	Paula's serial bytes both ways, modem lines, Paula's SERPER or the FIFO board's rate

4.1 Companion stream

TARGET	MODULE	COMMANDS USED
0	sysctrl.v	Machine settings (chipset, CPU, memory, region, IDE, turbo) and the battery clock; its interrupt is IRQ bit 2
1	hid.v	1 key {release, HID usage}; 2 mouse {buttons, dx, dy}; 3 joystick {device, fire 2, fire, up, down, left, right}
2	osd_u8g2.v	The core's OSD layer (unused by the firmware's setup screen, which draws on the heads)

4.2 Sector port

1. A floppy (drive bits 3:0) or IDE (bits 5:4) request raises SDC_REQ pending[31] with the drive mask and SDC_SECTOR; with sdc_hold the core's clock is held.
2. For a read the firmware writes 512 bytes into SDC_DATA (one byte a write, index from 0) from the floppy copy or the HD cache, then writes SDC_GO.
3. The core takes the buffer a byte every four core edges; for a write the core fills the buffer and sets done[30], and the firmware reads the 512 bytes back through SDC_DATA.
4. SDC_MOUNT with SDC_SIZE gives the core an image-mounted pulse for a drive: a disk change or a new hard disk.

The firmware fills and empties a block one byte a register access: about 0.78 ms a 512-byte block, 650 KB/s with nothing else running.

4.3 Blocks of HD0 answered by the firmware

BLOCKS	WINDOW	AMIGA PROGRAM
2	Mailbox: inhibit and release of a volume while the firmware writes into it	C:PANOFS
3	Command block: a console command and 16 KB of its output	ftdimon, flashmon
4–143	SHARE: requests, responses, 64 KB of file data	L:share-handler
144, 145, 256–383	Configuration flash: INFO, READ, WRITE, ERASE, CRC, PROGRAM with a 64 KB window	DEVS:panoflash.device

BLOCKS	WINDOW	AMIGA PROGRAM
511–895	Sockets: up to 20 operations a request on lwIP	C:PanoTCP (bsdsocket.library)

4.4 Interrupts

SOURCE	SERVICE
Audio FIFO watermark	Interrupt handler: drain, resample, refill the codec FIFO
Paula serial byte waiting	Interrupt handler: bytes to the USB-MIDI or FTDI OUT queue
USB interrupt completion (MIDI, FTDI)	Interrupt handler: IN data to Paula, OUT queue refilled
Millisecond timer	Time base; the main loop kicks the fabric watchdog
Sector requests, input, network, setup	Main loop (polled)

5 BOOT SEQUENCE

1. Power-up: the Spartan-6 multiboot header at flash 0x000000 loads the main slot at 0x380000 (the Amiga fabric); a configuration error falls back to the golden image at 0x040000.
2. The fabric's 32 KB boot RAM runs the block-RAM loader: it waits up to 3 s for DDR2 chip A, reads the flash image at 0x680000 (header magic and CRC32), copies it to 0x1000_0000 and starts it. A failure reconfigures to the golden image after 3 s.
3. That image, the stick loader, brings up USB and Ethernet, mounts the stick and checks \amiga\AMIGA.FW (magic, length, CRC32 of header and payload); it copies the firmware to 0x1000_0000 and starts it. With no stick or no valid file it stays at its network console.
4. The firmware reads \amiga\AMIGA.INI, loads the Kickstart image into chip B, reads the floppy images, opens the hard-disk images, sets the machine through sysctrl.v, waits up to NTPWAIT= seconds for network time, loads the battery clock and releases the core.

A firmware change is a copy of AMIGA.FW to the stick; the flash is written only for a new fabric or a new loader.

6 BUFFER SIZES

BUFFER	SIZE	PLACE
Chip-memory cache	64 KB, 2-way, 64-byte lines	Block RAM (core side)
Fast-RAM cache	32 KB, 2-way	Block RAM
CPU cache	4 KB instruction + 4 KB data, 2-way	Block RAM
Floppy images	2 MB a drive, 4 drives	DDR2 chip A
Hard-disk cache	8 MB: 128 × 64 KB, write-back, LRU	DDR2 chip A
Sector buffer	512 bytes	Fabric
Captured frames	3 × 1 MB (640 × up to 484 × 2 bytes)	DDR2 chip B
Capture line buffer	2 lines	Block RAM
Audio mix FIFO	1,024 stereo samples, watermark 256	Block RAM
Serial from Paula / to Paula	1,023 / 255 bytes	Block RAM
X-Surf receive queue	16 frames	Board model mailbox

BUFFER	SIZE	PLACE
USB DMA RAM	32 KB: control, 2 × 12 KB bulk, interrupt slots	Block RAM
Boot log	32 KB (newest lines), \amiga\BOOT.LOG	DDR2 chip A
Network console output ring	16 KB	DDR2 chip A
SHARE: data window	64 KB	HD0 blocks 16–143
Flash window	64 KB	HD0 blocks 256–383

7 MEASURED PERFORMANCE

MEASUREMENT	RESULT	CONDITIONS
CPU: two-instruction loop in fast RAM	254 ns an iteration (463 ns without cache)	membench under Workbench, 68020
CPU: loop in chip RAM	1,717 ns	as above
Fast RAM longword read / write	654 / 622 ns	as above
Chip RAM longword read	1,491 ns	as above
SysInfo 4	68020 at 41.60 MHz equivalent	A1200 profile, 8 MB fast RAM
HD0 through scsi.device	609 KB/s read, 620 KB/s write (32 KB)	hdbench
Stick read (USB bulk)	13.14 MB/s	16 MB, firmware benchmark
PanoTCP	329 KB/s receive, 375 KB/s send	2 MB, LAN
X-Surf (AmiTCP)	124 / 67 KB/s	1 MB
Serial, FIFO board	0 lost at 500,000 baud, about 27,000 bytes/s	FT232R, 5 s
MIDI loopback	0 lost, 0 errors at 100 % load	M-Audio 0763:1050, 3 × 10 s
Terminal (serialmon)	11,842 bytes/s at 80 × 25, smooth scrolling	vtplay, NTSC 640 × 200

8 IMPLEMENTATION

RESOURCE	USED	AVAILABLE	SHARE
Occupied slices	13,730	15,822	86 %
Slice LUTs	39,796	63,288	62 %
Slice registers	25,023	126,576	19 %
RAMB16BWER	176	268	65 %
RAMB8BWER	46	536	8 %
DSP48A1	34	180	18 %

CONSTRAINT	PERIOD	WORST SETUP SLACK
clk_sys	20.000 ns	0.334 ns
Video clock	12.500 ns	0.338 ns
ULPI	16.667 ns	0.511 ns
TG68K two-clock paths	40.000 ns	met

ISE 14.7 MAP and PAR of fabric 0x0AB5EF2D (G2_AMIGA_AGA, PAR cost table 7): all constraints met, timing score 0.

Register page from amiga/rtl/g2/amiga_g2.v and fw/amiga/amiga_regs.h; measurements on a G2 revision C unit. Repository mikewolak/pano-g2-amiga, release 1.0.